

# Replication Fork Slowing and Reversal upon DNA Damage Require PCNA Polyubiquitination and ZRANB3 DNA Translocase Activity

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## Evidence abstract

DNA damage tolerance during eukaryotic replication is orchestrated by PCNA ubiquitination.

While monoubiquitination activates mutagenic translesion synthesis, polyubiquitination activates an error-free pathway, elusive in mammals, enabling damage bypass by template switching.

Fork reversal is driven in vitro by multiple enzymes, including the DNA translocase ZRANB3, shown to bind polyubiquitinated PCNA.

However, whether this interaction promotes fork remodeling and template switching in vivo was unknown.

Here we show that damage-induced fork reversal in mammalian cells requires PCNA ubiquitination, UBC13, and K63-linked polyubiquitin chains, previously involved in error-free damage tolerance.

Fork reversal in vivo also requires ZRANB3 translocase activity and its interaction with polyubiquitinated PCNA, pinpointing ZRANB3 as a key effector of error-free DNA damage tolerance.

Mutations affecting fork reversal also induced unrestrained fork progression and chromosomal breakage, suggesting fork remodeling as a global fork slowing and protection mechanism.

Targeting these fork protection systems represents a promising strategy to potentiate cancer chemotherapy.

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